

SWAPNIL SINGH

+91 9115077085 | sswapnilsingh2002@gmail.com | GitHub | LinkedIn

EDUCATION

Indian Institute of Technology Madras - Chennai, India Current
Bachelor of Science in Data Science and Applications (Current CGPA: 9.41)

Rani Laxmi Bai Memorial School - Lucknow, India
High School (ICSE) - (Score: 85%)
Senior Secondary (CBSE) - (Score: 91.2%)
Achievements: Subject Topper (Computer Science)

SKILLS

- **Technical Skills:** Python, SQL, Flask, APIs, Excel, Google Colab, Git, GitHub, Object Relational Mapper (ORM), Asynchronous Programming in Python, Vue 2, Vuex, Asynchronous Programming in Vue 2, Matplotlib library, NumPy library, Pandas library, scikit-learn library, Redis Database
- **OS:** Windows and Linux
- **Core Competencies:** Database Management System (DBMS), Advanced Statistics, Hypothesis testing, Classification, Clustering, Regression, Principal Component Analysis (PCA), Kernel PCA
- **Courses:** Python - Prof. Sudarshan Iyengar, Machine Learning Foundation, Machine Learning Techniques - Prof. Arun Rajkumar
- **Languages:** Proficient in English and native Hindi

PROJECTS

Online Grocery Store (Level 1) - Used HTML, CSS and JavaScript for creating the frontend, used Python framework Flask and ORM to build APIs and other endpoints and database to store required details

Online Grocery Store (Level 2) - Multipage responsive web application, used HTML, CSS, and Vue 2 (JavaScript framework), used asynchronous programming and Python's SMTP library to send scheduled emails to customers, used caching, Redis database, token-based authentication, and database to store required details. GitHub Demo

Extra features: An accessibility menu is available for users who are color blind or have other visual impairments

Personal Projects

- **WordWise AI** - Engineered an interactive English language learning web application using Streamlit and Python to deliver comprehensive word lookups, etymology, and verb forms. Integrated the Google Gemini API to power generative, context-aware explanations, common phrase frequencies, and vocabulary enhancements. Implemented real-time audio playback and text-based phonetic stress guidance using the Google Text-to-Speech (gTTS) library. *Regularly using it from 1 year.* GitHub Live Website

EXPERIENCE & INTERNSHIP

Future Ready Talent Internship 90 days (Microsoft) - Web Application for NGO Based School

- Used HTML, CSS and JavaScript to create the frontend, used Azure Web app service to host the web application, and used Azure Blob Storage (for storing gallery photos)
- GitHub repository: <https://github.com/SWS-swapnilsingh/frtproject-ngowebapp>

Project TA - Modern Application Development 1 Project

- Provided mentorship to 17 students
- My mentees comprised 25% of all the submissions (All 11 active mentees successfully submitted their projects)
- Mentored and helped them resolve the Errors in their project

CERTIFICATIONS

- **Microsoft AI Challenge** - learned how to build a question answering bot, built a conversational understanding model, created a custom text classification solution, and created a text translator with the help of Azure AI Translator service
- **Azure Fundamentals** - learned about Azure storage, VMs, and other types of services provided by Azure
- **Code Chef Certificate** - 3 months long code chef python challenge, was able to come under 10% out of 700 students.
- **Azure AI Fundamentals** - learned about the fundamentals of question answering with the Language Service, fundamentals of machine learning, fundamentals of Azure AI Speech

LEADERSHIP, EXTRA-CURRICULAR & ACHIEVEMENTS

- **Reliance Foundation Scholar** - scholarship of rupees 2,00,000; among 5000 students selected from all over India
- **Group Leader of Corbett House (2022-2023)** - took 40+ Google Meet sessions, increased engagement of students
- **Multiple Awards and Certificates in Drawing Competition** - a certificate from Oil and Natural Gas Corporation (ONGC)
- **Participate in chess competitions and other sports** - in fests and school level